MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR (M.P.), INDIA
माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर (म.प्र.), भारत
(Deemed to be University)
NAAC ACCREDITED WITH A++ GRADE

Micro Project-II Report

On

Design & Development of Emotion Analyser

Subject Code: 29241210

Submitted For the Minor II Evaluation

Bachelor of Technology

In

Computer Science & Design

Submitted By

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Jan-June 2025

DECLARATION BY THE CANDIDATE

I hereby declare that the work entitled **"Design and Development of Emotional Analyser"** is my work, conducted under the supervision of **Dr. Dheeraj Kumar Dixit,** during the session Jan-Jun 2025. The report submitted by me is a record of Bona fide work carried out by me.

I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Date: 15th May 2025
Place: Gwalior

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Madhav Institute of Technology & Science, Gwalior, and the Department of Computer Science & Engineering for providing the necessary infrastructure, resources, and a supportive environment that enabled the successful completion of our project, "Design and Development of Emotion Analyser."

We are deeply thankful to Dr. R.K Pandit, Vice Chancellor, for his visionary leadership and continuous support.

Our heartfelt appreciation goes to Dr. Manjaree Pandit, Dean, Faculty of Engineering & Technology, for her encouragement and valuable insights throughout the project journey.

We are also grateful to Dr. Manish Dixit, Head of the Department, for his constant motivation and guidance.

Special thanks to our Project Guide, Dr. Dheeraj Kumar Dixit, whose expert supervision, constructive feedback, and consistent support played a pivotal role in shaping this work.

We also extend our thanks to Dr. Kirtiraj Bhatele, Project Coordinator, for his coordination, encouragement, and valuable suggestions.

We acknowledge the collaborative spirit and support of our peers and mentors, which greatly enriched our learning experience.

Lastly, we thank the authors of research papers, technical documentation, and the open-source community, whose contributions significantly supported the development and implementation of this project.

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1.Introduction

Emotions are a part of our everyday life — they influence the way we think, behave, and communicate with people. Whether a smile, frown, or surprise look, our feelings often show on our face without us saying a word. Emotion recognition is a new technology that assists machines in recognizing these emotions through learning facial expressions, tone of voice, and body language.

In simple terms, emotion recognition is the process of teaching computers how to recognize human emotions. This is done using artificial intelligence (AI) and machine learning. The system is trained on thousands of images and videos showing different emotional reactions. Over time, it learns patterns and becomes better at detecting how someone feels — whether they are happy, sad, angry, or relaxed.

This technology already finds its application in a great variety of domains. In medicine, it assists physicians in monitoring the state of mind of patients. In customer support and sales, it indicates how human subjects react to goods or commercials. It can even be employed for educational purposes to aid pupils' learning process, or for intelligent cars to control driver attention and tension levels.

2.Objectives

The primary goals of emotion recognition technology are:

1. Human Emotion Understanding – Recognize and comprehend emotions like happiness, sadness, fear, and surprise through AI.
2. Enhanced Human-Computer Interaction - Allow computers, robots, and virtual assistants to interact with users in a more natural and empathetic manner.
3. Personalized User Experience – Tailor content, ads, or suggestions to a user's emotional profile for more engaging interaction.
4. Security and Surveillance – Detect stress or suspicious activity in high security environments to avoid potential dangers.
5. Customer Feedback Analysis – Recognize customer feelings in stores, call centers, and internet services to enhance customer satisfaction.
6. Aiding Education and Training – Support teachers or AI instructors in identifying students' emotions in order to tailor teaching styles for enhanced learning experiences.

3. Work Flow

a) Data Collection

- The application will capture user dadta through a webcam (for facial recognition).
- The frontend (HTML+JavaScript) will access the webcam and send the data to the backend (Python Flask) for processing.
- Example: Using navigator.mediaDevices.getUserMedia() in JavaScript to access the camera and sending frames to a Flask endpoint via fetch() or WebSocket.

b) Processing of the Data

- The captured image (frame from webcam) is sent to the Python backend.
- In Python, image frames are processed using OpenCV to detect faces.
- Example: Using OpenCV in Python with Haar Cascade or DNN for face detection.

c) Feature Extraction

- For facial input, Python extracts facial landmarks (e.g., eyes, mouth, eyebrows) using mediapipe, dlib, or deep learning models.

d) Emotion Classification

- The extracted features are passed to a trained machine learning or deep learning model (e.g., TensorFlow, scikit-learn, or PyTorch) to predict emotions like happy, sad, angry, etc.
- The model runs in the Python backend and returns the predicted emotion to the frontend.
- Example: Use a pre-trained TensorFlow/Keras model in Python to classify emotions.

e) Display Results

- The frontend receives the emotion result via JSON response from Flask.
- Detected emotions are shown on the screen.

f) User Interaction

- The application can suggest actions based on the detected emotion.

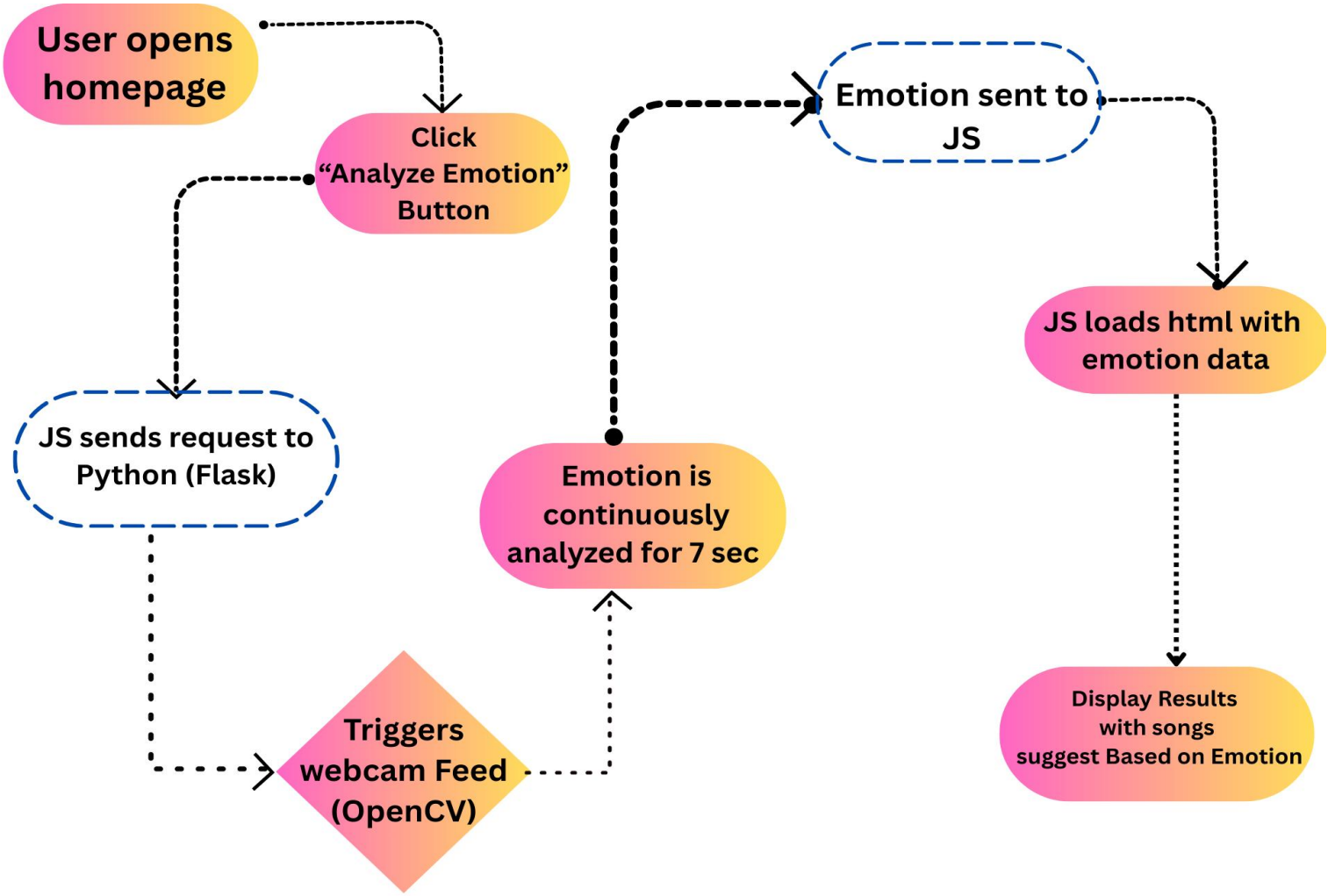


Fig. workflow of Emotion Analyser

4. Software and Hardware Requirements

1. Software Requirement

I. Frontend Technologies:

- HTML, CSS – For designing the user interface.
- JavaScript – For handling user interactions and real-time media capture.
  Libraries/Frameworks:
- WebRTC API – To access webcam via browser.
- Fetch API / Axios – To send data (image) to the Python backend.

II. Backend Technologies (Python + AI/ML Integration):

- Python (Flask) – For backend server, handling requests and responses.
- OpenCV / dlib / mediapipe – For face detection and feature extraction.
- TensorFlow / PyTorch / sc ikit-learn – To run emotion classification using trained ML/DL models.
- Pre-trained AI Models – For emotion classification based on facial expressions or voice.

III. Development Tools:

- VS Code – For writing frontend and backend code.
- Google Chrome / Firefox – For testing and debugging the web interface.
- Postman / cURL – To test and debug API requests and responses.
- Virtualenv – For managing Python environments and dependencies.

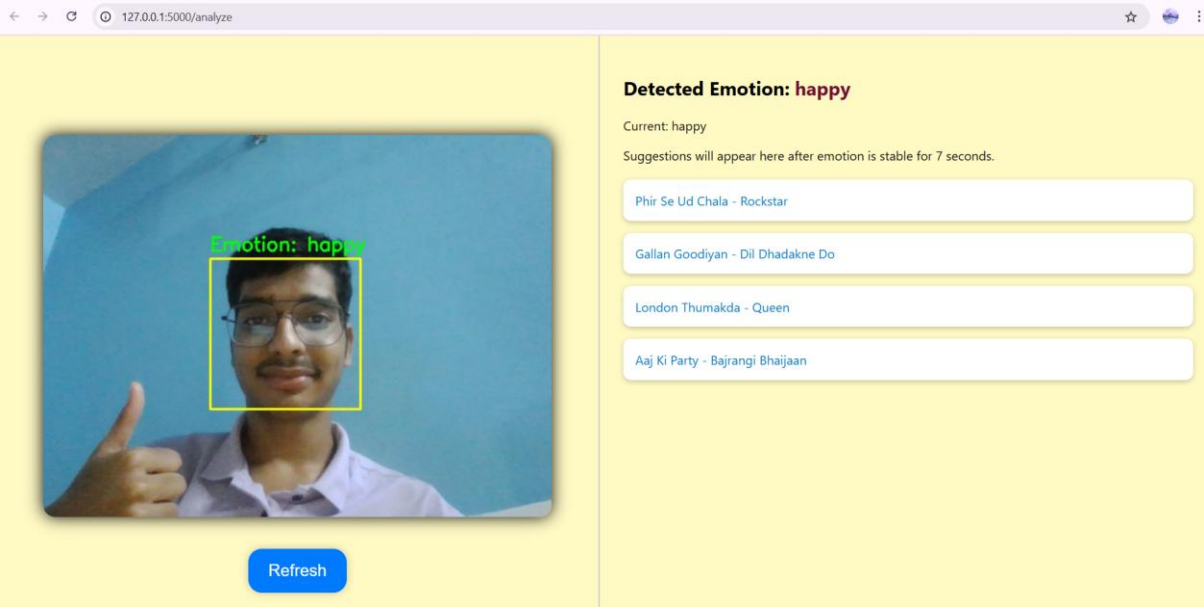
2.Hardware Requirement

Minimum System Requirements:

- Processor: Intel i5 or higher / AMD Ryzen 5 or higher.
- RAM: 8GB (16GB recommended for ML tasks).
- Storage: At least 10GB free space.
- GPU: Integrated GPU (for basic tasks) / Dedicated GPU (for deep learning).
- Camera: Webcam (for facial expression analysis).

5. Output

Emotion Analyzer
Detect your emotions and get personalized recommendations.
Analyze Emotion


127.0.0.1:5000/analyze
Emotion: happy
Refresh
Detected Emotion: happy
Current: happy
Suggestions will appear here after emotion is stable for 7 seconds.
Phir Se Ud Chala - Rockstar
Gallan Goodiyan - Dil Dhadakne Do
London Thumakda - Queen
Aaj Ki Party - Bajrangi Bhaijaan

6.Conclusion

This project brings together the power of modern web technologies (HTML, CSS, JavaScript) and Python-based backend development (Flask) to create a smart, emotion-aware system. By using real-time webcam input, the application detects users' emotions through advanced machine learning tools like OpenCV and TensorFlow. Once a stable emotion is recognized, the system responds intelligently suggesting mood-based content like songs .

Beyond just a technical showcase, this project highlights the growing role of emotion-aware computing in building more persona,l and intuitive digital experiences. It opens the door to applications that can respond empathetically to users whether in entertainment, healthcare, or education by making technology feel a little more human.